Mel HUAMAN MATOS

MECHANICAL ENGINEERING & ENERGY SYSTEMS ENGINEERING INTERN (AVAILABLE FROM APRIL 2026)

Prospective engineer and recipient of the 2025 National Innovation **First Prize from BRGM**. I combine strong technical expertise in thermodynamics, fluid mechanics, and energy systems with solid experience in managing complex projects. Focused on decarbonization and nuclear safety, I aim to contribute my industrial rigor and analytical capabilities to the Nuclear Department at Tractebel.

CONTACT DETAILS

+33 7 49 81 56 76

mel25mhmdr@gmail.com

mel-huaman-matos

Nantes, France

National Mobility

Work-study opportunity available

TECHNICAL SKILLS

- **Engineering:** Thermodynamics, Fluid Mechanics, Heat Transfer, Reliability Engineering, Life Cycle Assessment (LCA)
- **Software:** ANSYS, CATIA V5, ASPEN/HYSYS (advanced simulations), Excel VBA (Expert), Python

SOFT SKILLS

- Industrial rigor
- Multicultural teamwork
- Adaptability
- Technical communication

LANGUAGES

- French (B2)
- English (C1 – Advanced)
- Spanish (Native)
- Portuguese (A2)

STRENGTHS

- Rigor & Analytical Mindset
- Autonomy & Discipline
- Strong Interpersonal Skills
- Safety Awareness

INTERESTS

- Swimming (University Team Member)
- Dance
- Volunteer
  - Education Committee - 2025
  - Inspirate Girl - 2023
  - TEUNI - 2022

ACADEMIC BACKGROUND

IMT Atlantique, Nantes, France — **2025 - Present**

Master's-level General Engineering Degree (Double Degree Program)

- **Specialization:** Energy Systems and Low-Carbon Transition.
- **Key coursework:** Energy Networks, Energy Economics, Performance Management, Energy Risk & Management, Machine Energy Systems (Heat Exchangers and Engines), Decarbonized Production Technologies (Wind Power).
- **Project:** Design of the energy architecture for an innovative prototype.

Universidad Nacional de Ingeniería — Lima, Perú — **2020 - 2024**

Mechanical Engineering Degree (Top 5% of class)

- **Advanced coursework:** Advanced Thermodynamics II, Fluid Mechanics II, Heat and Mass Transfer, Engineering Economics and Finance.
- **Technical focus:** Turbomachinery, Industrial Maintenance, Reliability Engineering.
- **Research:** Achieved a 15% energy optimization in a high-efficiency system.

Université Nationale Autonome du Mexique — **2023**

Échange Académique – Mobilité d'Excellence

- Focus sur les énergies renouvelables et les véhicules décarbonés.

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern (Audit & Energy) — **2024 - 2025**

WSP — Lima, Perú

- **Cost Optimization:** Conducted energy audits and consumption analyses for critical infrastructure, identifying savings opportunities and supporting OPEX reduction.
- **Compliance & Standards:** Led cross-functional coordination to ensure full compliance with International Organization for Standardization standards (ISO 50001 – Energy, ISO 14001 – Environment, ISO 9001 – Quality).
- **Risk Management (HSE):** Assessed operational risks during the design phase and developed industrial safety protocols.

Project Manager & Technical Lead — **2023 - 2024**

Shell Eco-marathon

Full lifecycle design and development of a high-efficiency automotive prototype.

- **Energy Design:** Led the full design of the energy architecture for a high-efficiency electric prototype.
- **Modeling & Simulation:** Optimized energy consumption through aerodynamic and thermal simulations using Ansys.
- **Crisis Management:** Directed emergency interventions and resolved complex technical issues in field operations.

DISTINCTIONS & INNOVATIVE PROJECTS

BRGM (TInnov 2025) — Orléans, France — **2025**

Green hydrogen production from air-conditioning condensate

- **1st National Innovation Prize – TInnov 2025 (BRGM):** Designed and modeled a green hydrogen production system using condensate as the water source.

Challenge Convergence — Sète, France — **2025**

- **Energy Valorization:** Achieved a 9% technical performance improvement in a wave energy conversion system.

IMT Atlantique - Nantes , France — **2025**

Low-Power IoT System Development Project

- Designed and developed a low-power environmental monitoring module.

Research Project Competition — Lima, Perú — **2024**

Advanced modeling of hydrogen storage and distribution systems.

- **Innovation Award: Modeled a green hydrogen production system (TEUNI Project).**
- **H_2 Infrastructure: Actively contributed to the installation and optimization of hydrogen production systems.**